Traffy Fondue Insight: Where, When and What Bangkok Reports — Text and Spatial Mining of City Complaints

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Track Classification, clustering, association rules and prediction (text and spatial mining) · Rangsit University International College · 28 Oct 2026

1. Problem

Traffy Fondue is a LINE account where Bangkok residents report city problems (roads, flooding, lighting, rubbish). It holds over one million complaints [1]. Users choose the category themselves, so many are wrong, and **from November 2025 the category is empty for every complaint**. The city also has no clear view of **where** and **when** each problem happens most, which spots keep coming back, or where many complaints meet slow fixing (under 1 day to 269 days in our sample). Machine learning already sorts public complaints elsewhere [2]; on Traffy, past work sorts complaint text into categories [3][4] or predicts fixing time with one model [5]. Nobody combines text labelling with spatial and time mining to show where and when to focus.

2. Research question

Where, when and what kinds of problems do Bangkok residents report most on Traffy Fondue, and which areas and times should the city focus on, using complaint types labelled from text (including translated text)?

- **RQ1 What:** which model labels the problem type from text best (also after Thai–English translation), fixing wrong and missing categories?
- **RQ2 Where:** which districts and exact spots are hotspots per problem type (per 10,000 people), and which spots keep coming back?
- **RQ3 When and why:** which season, month, weekday and hour have the most complaints, and are flooding and road complaints linked with heavy rain?
- **RQ4 Focus:** which place + time + problem go together, which hotspots are frequent and slow, and where should the city focus next week?

3. Dataset (confirmed)

Source	Traffy Fondue open data, Bangkok Metropolitan Administration. publicapi.traffy.in.th/dump-csv-chadchart/bangkok_traffy.csv
Real / size	Real complaints, over 1 million rows (about 25,000–35,000 per month). Far above the 500-row minimum.
Period	Jun 2022 – Oct 2025 = labelled training data (empty category: 9.8% in 2023, 10.8% in 2024). Nov 2025 – 2026 has no category, so our text model labels it.
Columns	comment (text), type (labels), coords (GPS), district, subdistrict, organization, timestamp, last_activity, state, count_reopen. All 16 columns checked.
Extra data	Daily Bangkok rainfall and district population, joined by date and district. Seasons: hot Mar–May, rainy Jun–Oct, cool Nov–Feb.

4. Method (CRISP-DM, all four data mining tracks)

Track	Task	Baseline and compared methods	Evaluation
Classification	Problem type from text	TF-IDF + Logistic Regression (baseline); Linear SVM, Random Forest, WangchanBERTa (vs. Traffy model [4]); Thai vs. translated text	Micro/macro F1, Hamming loss, 5-fold
Clustering	Hotspots (where)	DBSCAN, HDBSCAN on GPS; kernel density; Getis-Ord Gi*; k-means on district profiles	Silhouette, hotspot significance
Association rules	Place + time + problem	Apriori, FP-Growth on {district, season, hour band, rain} → type	Support, confidence, lift
Prediction	Next-week load	Last-week baseline vs. XGBoost per district and type; priority = frequency × slowness	MAE; train 2022–24, test 2025

5. Changes after Milestone 1 and expected contribution

Following Ajarn's feedback, **text labelling comes first**, translation lets our team (who cannot read Thai) check results, and an **interactive monitoring app** shows hotspots by area, type, season and hour, a priority list and a next-week forecast. Contribution: labels for 2025–2026 complaints that have none, fair (per-person) hotspot maps, rain-linked seasonal patterns and a clear "where and when to focus" list for the city. Limits: report time is not event time, and busy areas may report more.

6. Next milestones

M3 (10/12 Nov): data cleaned, rainfall and population joined, text baseline + 2 models (mean ± std), first hotspot map and season chart.
M4 (24/26 Nov): rules, priority map, forecast, app, IEEE paper (10+ refs), 10-min talk.

[1] BMA PR Office, "Bangkok Celebrates Digital Milestone as Traffy Fondue Resolves One Million City Complaints," Mar. 2026.
[2] F. Caldeira, L. Nunes, R. Ribeiro, "Classification of Public Administration Complaints," SLATE 2022, OASIcs vol. 104, 2022.
[3] A. Pongsathornwiwat, P. Kaewvichit, "Developing Multi-Label Classification Model for Improving Text Categorizing Problems a Case of Traffy* Fondue Platform," Sci. & Tech. Asia, 29(2), 2024.
[4] KDAI-NLP, "wangchanberta-traffy-multi," Hugging Face model. [5] karnjj, NonKhuna et al., "Sed-Young-Wa," GitHub repository, 2023.